

Total No. of Questions : 8]

SEAT No. :

P2325

[Total No. of Pages : 2

[5870]-1122

T.E. (Chemical)

PROCESS INSTRUMENTATION AND CONTROL

(2019 Pattern) (Semester - II) (Elective - II) (Theory) (309351B)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q. 1 or Q. 2, Q.3 or Q. 4, Q.5 or Q. 6, Q. 7 or Q. 8.
- 2) Figures to the right indicate full marks.
- 3) Neat diagram must be drawn wherever necessary.
- 4) Use of logarithmic tables, slide rule, Mollier charts, electronic p and steam table is allowed.
- 5) Assume suitable data, if necessary.

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Q1) a) Explain with diagram, construction and working of bellows. **[8]**

b) What are transducers? Explain types of transducers. **[8]**

OR

Q2) a) Enlist various pressure measuring instruments and explain U tube manometer in detail. **[8]**

b) Explain with diagram, construction, working and calibration of pressure sensor using dead weight tester. **[8]**

Q3) a) Explain with diagram, construction and working of Orifice meter with its industrial application. **[10]**

b) How level can be measured using radioactive transducers? Draw neat sketch and explain in detail. **[8]**

OR

Q4) Write short note on : **[18]**

a) Radiation level method

b) Sight glass method

P.T.O.

Q5) a) Explain principle with diagram, construction and working of pH meter. [8]

b) Write short note on : [10]

i) Refractometry

ii) Liquid chromatography

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OR

Q6) Describe with neat diagram the following techniques of composition analysis. [18]

a) Gas chromatography

b) Mass spectroscopy

Q7) a) Derive the transfer function of mercury in glass thermometer and find the dynamic behavior of step change in input. [9]

b) Give classification of process variable with respect to process control. [9]

OR

Q8) a) Explain with equation, different control actions. [9]

b) What are servo and regulatory operation? [9]

